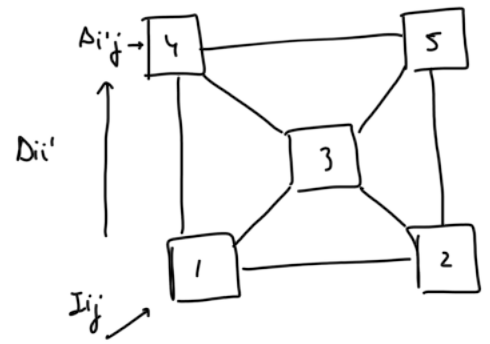


# Data center problem

## • SETS

$i: \{1, \dots, 10\}$  node number

$j: \{1, \dots, 80\}$  task number



## • PARAMETERS

$C_i$ : Maximum capacity of node  $i$  [Mb]

$S_j$ : Memory required by task  $j$  [Mb]

$I_{ij}$ : If  $i$  is the initial node of task  $j$   $\{0, 1\}$

$T_j$ : Time to complete task  $j$  [s]

$D_{ii'}$ : Time to transport a task from node  $i$  to  $i'$  [s]

$P_j$ : Priority group of task  $j$   $\{1, 2, 3\}$

$Q_i$ : Quality group of node  $i$   $\{1, \dots, 10\}$

$\varepsilon$ : Energy decrease required [%]

$R_i$ : Time increase due to reduction in processing capacity [s/%]

## • VARIABLES

$X_{iij}$ : If task  $j$  is transported from node  $i$  to  $i'$   $\{0, 1\}$

$A_{ij}$ : If task  $j$  is assigned to be completed in node  $i$   $\{0, 1\}$

$y_i$ : % of energy reduced in node  $i$

$u_i$ : Variable for extra time delay if  $y_i > 20$

$z$ : Minimum satisfaction for tasks

$EX_i$ : Auxiliar binary variable that indicates if  $y_i > 20$   $\{0, 1\}$

$M_{jj'}$ : If task  $j$  has priority over task  $j'$   $\{0, 1\}$

## • OBJECTIVE FUNCTION

$$a) \min \sum_{i, i', j} (T_j + X_{iij} \cdot D_{ii'} + (y_i + u_i) \cdot R_i)$$

$$b) \max z$$

## • CONSTRAINTS

- Each task is not transported more than once

$$\sum_{i,i'} X_{i'i,j} \leq 1 \quad \forall j$$

- Each task is assigned to a single node

$$\sum_i \Delta_{ij} = 1 \quad \forall j$$

- Maximum processing capacity for each node

$$\sum_j \Delta_{ij} \cdot S_j \leq C_i \cdot (100 - y_i) / 100 \quad \forall i$$

- Task assigned to a different node means it has to be transported

$$I_{ij} \cdot \Delta_{i'j} = X_{i'i,j} \quad \forall i, i', j \quad i \neq i'$$

- More important tasks are assigned to better nodes

$$\left[ \text{IF } P_{j'} - P_j \geq 1 \rightarrow \Pi_{jj'} = 1 ; P_{j'} - P_j \leq 1 - 1 + (M+1) \Pi_{jj'} \quad 3 \leq n+1 ; n \geq 2 ; n=2 \right]$$

$$P_{j'} - P_j \leq 3 \Pi_{jj'} \quad \forall j, j' \quad j \neq j'$$

$$\left[ \text{IF } \Pi_{jj'} = 1 \rightarrow \sum_i (A_{ij'} - A_{ij}) \cdot Q_i \geq 0 ; \sum_i (A_{ij'} - A_{ij}) Q_i \geq m(1 - \Pi_{jj'}) \right]$$

$$-10 \geq m ; \underline{m = -10}$$

$$\sum_i (A_{ij'} - A_{ij}) Q_i \geq -10(1 - \Pi_{jj'}) \quad \forall j, j' \quad j \neq j'$$

$$\Pi_{jj'} + \Pi_{j'j} \leq 1$$

- Energy decrease must be satisfied

$$\sum_i y_i \cdot C_i \geq \varepsilon \cdot \sum_i C_i$$

- $y_i$  is a percentage

$$y_i \geq 0 \quad \forall i$$

$$y_i \leq 100 \quad \forall i$$

- Extra time if a node is under 80% of its capacity

$$\left[ \text{IF } y_i - 20 \geq 0 \rightarrow \varepsilon x_i = 1 \quad y_i - 20 \leq -1 + (M+1) \varepsilon x_i \quad 80 \leq n ; n=80 \right]$$

$$\frac{1}{81} (y_i - 19) \leq \varepsilon x_i \quad \forall i$$

$$\left[ \text{IF } \varepsilon x_i = 1 \rightarrow y_i - u_i \leq 0 \right]$$

$$y_i - u_i \leq n(1 - \varepsilon x_i) \quad 20 \leq n ; n=20$$

$$u_i \geq y_i - 20(1 - \varepsilon x_i) \quad \forall i$$

$$u_i \geq 0 \quad \forall i$$

- Register minimum satisfaction

$$2 \leq \sum_i A_{ij} \cdot \frac{1}{Q_i} \cdot \frac{1}{P_j} \quad \forall j$$

1: Mas importante

1: Mejor nodo